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February 5, 1980
SUPERSEDING
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April 6, 1967

FEDERAL SPECIFICATION

HEATING ELEMENTS, ELECTRICAL (REPLACEMENTS FOR DOMESTIC APPLIANCES)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers electrical heating elements used as replacement items for water heaters, ranges, dishwashers, dryers, and hotplates.

1.2 Classification.

1.2.1 Types and styles. Heating elements shall be of the following types and styles, as specified (see 6.2).

Type I - Immersion units.

Style A - Flange mounted.
Style B - Screw plug mounted.

Type II - Non-immersion units.

Style A - Coiled surface units.
Style B - Center mounted loop.
Style C - U-shaped loop.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

H28 - Screw-Thread Standards for Federal Services.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standard, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

FSC 4540

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Underwriters' Laboratories, Inc. (UL):

UL1030 - Sheathed Heating Elements.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a heating element for first article inspection and approval (see 4.3.1 and 6.4).

3.2 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

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3.3 Identical items. Heating elements of the same classification furnished under any specific contract shall by physically and mechanically identical. This requirement includes parts, assemblies, components, and accessories. No deviation will be acceptable without prior written approval of the contracting officer.

3.4 Interchangeability. All heating elements shall be manufactured to standards that will permit replacement without modification of the element or the equipment in which it is installed. All threaded parts shall conform to FED. STD. H28.

3.5 Electrical characteristics. The heating element shall be designed to operate on 60 Hertz alternating current (ac) at the voltage and wattage specified (see 6.2). Voltage shall be as follows:

Voltage	Phase
120 volts	Single
120/240 volts	Single
240 volts	Single
208 volts	Single/3 phase

The type of switch used with surface units will determine the voltage:

2, 3, or 4 heat switch	120 volt unit
5 or 7 heat switch	208/240 volt unit

3.6 Construction. The heating element shall consist of straight or helically coiled nickel-chromium, or iron-chromium-aluminum-colbalt alloy wire mounted centrally within the tubular or cast metal sheath. Other resistance wire alloys of equivalent performance may be used when approved by the contracting officer or agency concerned. The space between the coil and sheath shall be solidly packed with magnesium oxide material, and in accordance with UL1030. The ends of elements shall contain a moisture barrier to retard the entrance of water or vapors and thereby preclude a short circuit. Terminals shall be gas shield arc, projection or spot welded to the resistor element to insure adequate strength and electrical conductivity.

3.6.1 Immersion elements. The sheath material shall be copper tubing, or when specified (see 6.2), electroplated. The heating portion of the element shall be the single or fold-back loop shape, as specified (see 6.2). Unless otherwise specified (see 6.2), a suitable flange gasket shall be provided for style A elements. Style B elements shall be provided with the National Pipe Thread size specified (see 6.2). Corrosion-resistant steel, style B, elements providing moisture sealed terminals shall be supplied when specified (see 6.2). When specified (see 6.2), a thermowell shall be a part of the immersion elements. When specified (see 6.2), a thermostatic control shall be provided with each immersion element. The desired temperature range of heating element shall be as specified (see 6.2).

3.6.2 Non-immersion elements. The element sheath shall be nickel alloy, corrosion-resistant steel tubing. The electrical connections shall be made by a screw type or standard threaded stud type fastener attached to the ends of the heating element or by a plug consisting of three prongs on 7/8-inch centers. All others will be as specified (see 6.2).

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3.6.3 Coiled surface units (style A). The nickel-chromium-iron alloy sheath shall be formed to provide a flattened top surface to increase the conductive surface. The width of the flattened tube shall be at least 75 percent of the nominal tube diameter. A corrosion-resistant steel-supporting assembly providing not less than three point support, 120° apart, shall be fastened to the coil by crimping or by use of retainer cups welded to the element. The nominal diameter of element shall be 6 inches corresponding to an appliance opening of 6-1/2 to 9 inches in diameter, or 8 inches corresponding to an appliance opening of 8-1/2 to 10-1/2 inches in diameter.

3.6.4 Center mounted (style B) and U-shaped (style C) loop elements. Elements shall be of the dimensions specified or shall be compatible with the oven dimensions specified (see 6.2). Elements used in ovens for baking shall provide a minimum of 1-inch clearance from the oven sides and 2-inch clearance from its front. Unless otherwise specified (see 6.2), two supporting brackets shall be supplied with the baking element. Elements used for broiling shall provide 1/2-inch minimum clearance from the front. Elements used for dishwashers shall be provided with a suitable water-tight sealing device to insure zero-leakage when the unit is installed.

3.7 Grounding. Provision shall be made for grounding each heating element to the appliance. The grounding prong on plug-in elements shall extend a minimum of 1/8-inch beyond the other prongs. The mounting screws of immersion heaters shall be considered an adequate ground.

3.8 Accessories. The following items shall be shipped as component parts when specified (see 6.2).

3.8.1 Adapter rings. Adapter rings for range and hotplate surface units shall be made of chrome-plated steel, or stainless steel, with sizes as specified (see 6.2).

3.8.2 Drip pans. Drip pans for range and hotplate surface units shall be made of chrome-plated, stainless, vitreous enameled steel, or aluminum, with sizes as specified (see 6.2).

3.8.3 Plug-in blocks. Range bake and broil element terminal blocks shall be made of ceramic material with permanently mounted blocks shall be made of ceramic material with permanently mounted connecting lugs. Unless specified (see 6.2), the block shall accept elements with three prongs on 7/8-inch centers.

3.9 Workmanship. The quality of workmanship shall meet the standards prevalent among manufacturers who normally produce equipment of the type specified herein.

3.10 Instructions. Each heating element ordered shall have installation instructions pertaining to that element. Safety instructions shall also be printed along with the installation instructions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for

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the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection lot. All elements of the same size, type, and style offered to the Government under a specific contract, shall be considered a lot for purposes of inspection.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3.1).
(b) Quality conformance inspection (see 4.3.2).
(c) Preparation for delivery inspection (see 4.7).

4.3.1 First article inspection. First article inspection shall be performed on one heating element when a first article sample is required (see 3.1 and 6.2). This inspection shall include the examination of 4.5 and the tests of 4.6. Unless otherwise specified (see 6.4) first article may be a standard production item from the contractor's current inventory provided the heating element meets the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining heating elements to be furnished under the contract.

4.3.2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.5 on samples selected in accordance with 4.4.2. Any unit in a sample containing one or more defects shall be rejected. The lot represented by the sample shall be accepted or rejected in accordance with the quality acceptance level of 4.4.1 and 4.4.2.

4.4 Sampling.

4.4.1 Sampling for examination. A random sample of elements shall be selected from each lot in accordance with MIL-STD-105 at inspection level II. In terms of defects per 100 units, the Acceptable Quality Level (AQL) shall be 1.5 for major defects and 4.0 for minor defects.

4.4.2 Sampling for tests. A random sample of elements shall be selected from each lot offered to the Government in accordance with MIL-STD-105 at inspection level II. The AQL shall be based on 1.5 percent defective units.

4.5 Examination. Each element shall be examined for compliance with the requirements specified in section 3 of this specification. Examination shall be conducted as specified in table I. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

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TABLE I. Classification of defects.

Categories	Defects
Critical:	None defined.
Major:	
101	Construction not as specified.
102	Electrical characteristics not as specified.
103	Wrong sheathing material.
104	Unit has parts missing.
Minor:	
201	Immersion elements not sufficiently tin plated.
202	Ground marking missing.
203	Accessories wrong material or missing.
204	Workmanship not of high quality

4.6 Tests. Each sample selected in accordance with 4.4.2 and the first article shall be tested in accordance with 4.6.1, 4.6.2, and 4.6.3. Any unit failing to pass the following test, as applicable, shall be rejected. If the number of rejected units in any sample is equal to or greater than the rejection number for that sample, the lot represented by the sample shall be rejected. Certified test reports from an independent laboratory approved by the contracting officer will be accepted as evidence that the unit has passed the required tests.

4.6.1 Dielectric test. Each sample shall be tested by supplying 1000 volts, 60 cycle, single phase ac for 1 minute between the exposed non-current carrying parts, with the unit at maximum operating temperature attained in normal use. The method of testing will be in accordance with UL1030, then a visual examination for arcing or material breakdown shall be made during the 1-minute interval at set conditions. Any arcing or material breakdown shall be cause for rejection. When the examination is completed, the voltage should be gradually reduced to zero.

4.6.2 Normal operation test. The element shall be mounted to simulate appliance installation. Provisions shall be made to measure input voltage, wattage, and element surface temperature. With use of a suitable controlling device, the element shall be operated at maximum conditions for 1 hour. During this period, the wattage shall not fall below 95 percent of its rated value. Failure to meet this condition shall be cause for rejection. Coiled surface units (type II, style A) shall be required to attain 90 percent of the maximum operating surface temperature within 1 minute. Failure to meet this condition shall be cause for rejection.

4.6.3 Distortion test. Coiled surface units (type II, style A) shall be mounted to simulate appliance installation. The element shall be brought to maximum operating condition, which shall be maintained for a period of 1 hour. An 8 pound weighted straight edge shall be laid across the element. A feeler gage shall be used to measure the gap between the straight edge and the heating surface. Check gap before and after heat test. A distortion of 0.032 inch or more shall be cause for rejection.

4.7 Preparation for delivery inspection. The preservation, packing, and marking of the heating elements shall be inspected to determine conformance to the applicable requirements of section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packing, and marking. Unless otherwise specified (see 6.2), the heating elements shall be preserved and packed in commercial quantities in accordance with the supplier's standard practice for domestic or overseas shipment. The preservation and packing shall provide protection from deterioration and damage and shall, as a minimum, meet the applicable carrier rules and regulations. Marking shall be as specified in the contract.

5.1.1 Military procurements. When specified (see 6.2), the heating elements shall be preserved, packed, and marked in accordance with MIL-E-17555 with the degree of protection specified.

6. NOTES

6.1 Intended use. This specification covers immersion-type electrical heating elements for use in domestic hot water heaters; tubular-type electrical heating elements for use in domestic ranges, hotplates, dishwashers, and dryers.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and style of element (see 1.2.1).
(c) When a first article sample is required for inspection and approval (see 3.1, 4.3.1, and 6.4).
(d) Wattage required (see 3.5), (when making a substitute, do not use more than a 500 watt larger unit).
(e) Voltage required and whether single or 3 phase (see 3.5).
(f) For immersion elements (see 3.6.1), specify:

(1) Length of loop.
(2) Shape - single or folded loop.
(3) Bolt pattern on style A elements.
(4) If gasket on style A element is not required.
(5) Pipe thread size on style B elements.
(6) With or without thermowell on style B elements.
(7) With or without thermostatic control mounted on flange assembly.
(8) When heating elements shall be electro-plated.
(9) When corrosion-resistant steel elements are required.
(10) Temperature range required.

(g) For non-immersion elements.

(1) Plug-in units - measure distance between prong centerlines. If nonstandard (see 3.6.2), give sketch of dimensions.
(2) Surface units - indicate diameter of the appliance opening and nominal element diameter (see 3.6.3).
(3) For style B and C elements - indicate length and width of the oven bottom or give dimensions of the elements (see 3.6.4).
(4) When supporting brackets are not required with baking element (see 3.6.4).

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(h) When adapter rings and drip pans for surface units are shipped as component parts, specify element diameter (see 3.8.1 and 3.8.2), and appliance opening.
(i) When plug-in blocks are to be shipped as component parts (see 3.8.3) and if block must accept other than three prongs on 7/8-inch center provide the following:

(1) Spacing of prong holes.
(2) Overall size.
(3) Mounting requirements.

(j) When heating elements are to be preserved, packed, and marked in accordance with MIL-E-17555 (see 5.1).
(k) Marking required (see 5.1.1).

6.3 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article is required, it shall be tested and approved under the approriate provisions of paragraph 7-104.55 of the DAR. The first article should be a first production item consisting of one complete heating element or it may be a standard production unit from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instruction in all procurement instruments regarding arrangements for examinations, tests, and approval of the first article.

6.5 Civil agency requirements. When level B packaging is required for civil agency procurements, the level A requirements in 5.1.1 shall apply.

Military custodians:

Navy - YD
Air Force - 99

User activity:

Navy - MC

Civil Agency Coordinating Activity:

FEC - HEW

Preparing activity:

Navy - YD

Project No. 4540-0054

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